LAB MANUAL

Subject: Statistical Computing Techniques using R Lab

Subject Code: BTCS 502-20

Class: B. Tech 5th Semester

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2.	Perform simple arithmetic's using R.
3.	Perform basic R functions.
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5.	Create different charts for visualization of given set of data.
6.	Find the mean, median, standard deviation and quartiles of a set of observations.
7.	Find the Skewness and Kurtosis of a given dataset distribution.
8.	Given the scenario, implement the Bayes rule by finding the posterior probability.
9.	Find the mass function of a binomial distribution with n=20,p=0.4n=20,p=0.4. Also draw the graphs of the mass function and cumulative distribution function.
10.	Generate and draw the cdf and pdf of a normal distribution with mean=10 and standard deviation=3. Use values of xx from 0 to 20 in intervals of 1.
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INSTITUTE VISION

- To emerge as an institution of technical excellence imparting professional education for sustainable development of society.

INSTITUTE MISSION

- To provide quality technical education through state-of-the-art infrastructure and well qualified and experienced faculty.
- Having academic flexibility through strong industry academia interactions.
- Focus on students' employability, entrepreneurship, higher education and competitive examinations.
- To Inculcate ethical and moral values in students.

DEPARTMENT VISION

- To provide imperative skills to students to meet industry needs and become responsible engineers, entrepreneurs, and citizens.

DEPARTMENT MISSION

- To educate the students in the field of computer science with ever-changing technologies and skills.
- To enable the students in solving real-time problems and make use of new technologies.
- To have industry collaboration and interaction with professional societies for continuous development.
- To help students in becoming successful entrepreneurs.

PROGRAMME EDUCATIONAL OBJECTIVES (PEOs)

PEO1: To develop ethical computer engineers, administrators, and business leaders, graduates in computer engineering must cultivate a range of skills and leaders.

PEO2: To prepare professionals who assess software requirements and deliver solutions through impactful product designs, innovative thinking, analysis, and decision- making.

PEO3: To empower graduates with lifelong skills in data analysis, design, and synthesis, enabling them to continuously develop innovative products and solutions that meet evolving industrial demands.

PEO4: To foster a mindset for cultivating a professional and ethical demeanor, and adeptly engaging with global challenges, thereby effecting positive impacts on the environment and society.

PROGRAM SPECIFIC OUTCOMES (PSOs)

After the completion of the course, B. Tech Computer Science and Engineering, the graduates will have the following Program Specific Outcomes:

- **PSO1:** To use principles of Computer Science & Engineering (such as software engineering, computer networks, data structures &computer programming high level languages) for developing software solutions.
- **PSO2:** To clearly understand the concept of programming languages, computer architecture and their applications in different field of technologies to develop cost-effective solutions in the area of computer science by the use of various methodological algorithms and different tools.

PROGRAM OUTCOMES (POs)

Engineering Graduates should possess the following:

1. **Engineering knowledge**: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

2. **Problem analysis**: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

3. **Design / development of solutions**: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

4. **Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

5. **Modern tool usage**: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

6. **The engineer and society**: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

7. **Environment and sustainability**: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

8. **Ethics**: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

9. **Individual and team work**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

10. **Communication**: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

11. **Project management and finance**: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multi-disciplinary environments.

12. **Life- long learning**: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Experiment 1

Experiment name: Familiarization with R Environments

Objectives: To understand the various environments in R and their functionalities.

Prerequisites: Basic knowledge of programming concepts, understanding of data analysis workflow.

Key terms: R Console, RStudio, Workspace.

Experimental set-up/Equipment/Apparatus/Tools: -

1. Computer System
2. R and RStudio installed.

Theory and application

R offers a versatile platform for statistical computing and graphics. Familiarity with the RStudio environment, including the Console, Editor, and Workspace tabs, is crucial for smooth programming.

Experimental Procedure-

1. **Install and load a package**:

```
# Install a package
install.packages("dplyr")
# Load the package
library(dplyr)
```

2. **Explore environment and variables**:

```
# Create variables
x <- 10
y <- 20
result <- x + y

# Check environment
ls()

# Print result
print(result)
```

3. **Basic mathematical operations**:

```
sum <- 5 + 3
product <- 6 * 7
sqrt_value <- sqrt(25) # Print outputs
sum
product
```

sqrt_value

Precaution and sources of error:

- Ensure the versions of R and RStudio are compatible.
- Save the workspace regularly to prevent data loss.

Results

Expected Output:

```csharp
[1] "dplyr"
[1] 30
[1] 8
[1] 42
[1] 5
```

Conclusions

Familiarizing with R environments supports smooth workflow and programming.

Related short question:

1. What is the role of the R Console?
2. Differentiate between R and RStudio.
3. How can you save your current workspace in RStudio?

Experiment 2

Experiment name: Perform Simple Arithmetic Using R

Objectives: To execute basic arithmetic operations in R.

Prerequisites: Knowledge of arithmetic operations, familiarity with basic syntax.

Key terms: Operators, modulus (%%), power (^).

Experimental set-up/Equipment/Apparatus/Tools: -

1. Computer System
2. R and RStudio installed.

Theory and application

R supports basic arithmetic operations such as addition, subtraction, multiplication, division, modulus, and exponentiation, allowing seamless numerical computations.

Experimental Procedure-

```
# Simple arithmetic operations in R

addition <- 15 + 10

subtraction <- 30 - 12

multiplication <- 7 * 5

division <- 100 / 25

# Modulus and exponentiation

modulus <- 15 %% 4

exponentiation <- 2^4

# Print all results

addition

subtraction

multiplication

division
```

modulus

exponentiation

Precaution and sources of error:

- Use correct operator syntax to avoid errors.

Results

Expected Output:

```csharp
[1] 25
[1] 18
[1] 35
[1] 4
[1] 3
[1] 16
```

Conclusions

Performing arithmetic operations is essential for numerical analysis and data manipulation in R.

Related short question:

1. What is the syntax for exponentiation in R?
2. How do you compute the remainder of a division in R?
3. Can R perform arithmetic operations on vectors?

Experiment 3

Experiment name: Perform Basic R Functions

Objectives: To utilize and understand fundamental R functions like print(), mean(), and sum().

Prerequisites: Basic understanding of function usage, familiarity with syntax.

Key terms: Functions, parameters, arguments.

Experimental set-up/Equipment/Apparatus/Tools: -

1. Computer System
2. R and RStudio installed.

Theory and application

Functions in R allow efficient computation and simplify complex tasks. Common functions include mean()
for averages and sum() for cumulative totals.

Experimental Procedure-

```
# Create a vector of numbers

numbers <- c(10, 20, 30, 40, 50, 60)

# Find basic statistics

mean_value <- mean(numbers)

median_value <- median(numbers)

sd_value <- sd(numbers)

var_value <- var(numbers)

# Print the results

mean_value

median_value

sd_value

var_value
```

Precaution and sources of error:

- Ensure input data is valid to avoid runtime errors.

Results

Expected Output:

```csharp
[1] 35
[1] 35
[1] 18.70829
[1] 350
```

Conclusions

Basic R functions streamline computations and improve code readability.

Related short question:

1. What is the difference between print() and cat() in R?
2. How do you calculate the mean of a numeric vector in R?
3. Why is error-checking important when using R functions?

Experiment 4

Experiment name: Use Graphical Techniques in Exploratory Data Analysis (EDA)

Objectives: To explore datasets visually using EDA techniques.

Prerequisites: Knowledge of EDA concepts, basic understanding of data visualization.

Key terms: Histogram, boxplot, scatter plot.

Experimental set-up/Equipment/Apparatus/Tools: -

1. Computer System
2. R and RStudio installed.
3. Required libraries (like ggplot2) installed.

Theory and application

EDA leverages visual tools to identify patterns, trends, and outliers. Graphical tools such as histograms and
boxplots provide quick insights into the data.

Experimental Procedure-

```
# Create a dataset
data <- data.frame(
  x = c(1, 2, 3, 4, 5),
  y = c(2, 4, 6, 8, 10)
)
# Scatter plot
plot(data$x, data$y, main="Scatter Plot", xlab="X-axis", ylab="Y-axis")
# Line plot
plot(data$x, data$y, type="l", col="blue", main="Line Plot")
# Bar plot
barplot(data$y, names.arg=data$x, col="green", main="Bar Plot")
```

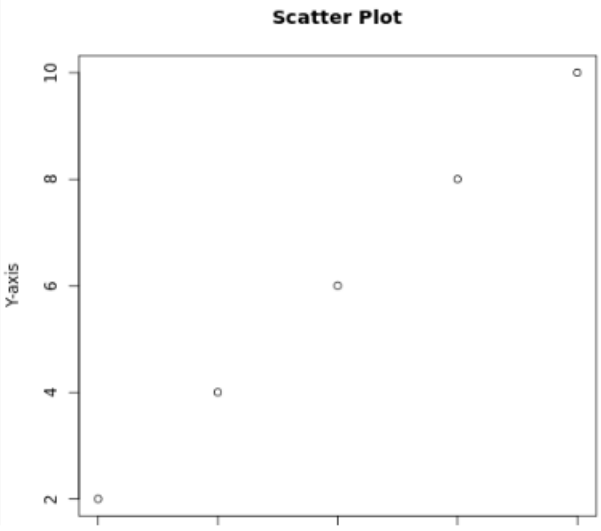
Precaution and sources of error:

- Use appropriate scaling and proper axis labels for clear visualization.

Results

Output

```
[Execution complete with exit code 0]
```



Conclusions

EDA with graphical tools enhances data interpretation and decision-making.

Related short question:

1. What type of data is best suited for a boxplot?
2. How do you plot a histogram in R?
3. Why is labeling important in visualizations?

Experiment 5

Experiment name: Create Different Charts for Data Visualization

Objectives: To create bar charts, pie charts, and line charts for data visualization.

Prerequisites: Familiarity with chart types, basic understanding of data organization

Key terms: Bar chart, pie chart, line chart.

Experimental set-up/Equipment/Apparatus/Tools: -

4. Computer System
5. R and RStudio installed.
6. Required libraries (like ggplot2) installed.

Theory and application

Different charts serve specific purposes in data visualization. Bar charts compare categorical data,

pie charts show proportions, and line charts depict trends over time.

Experimental Procedure-

```
# Example data

values <- c(5, 10, 15, 20, 25)

# Pie chart

pie(values, labels=c("A", "B", "C", "D", "E"), main="Pie Chart Example")

# Bar plot

barplot(values, col=rainbow(length(values)), main="Bar Plot Example")
```

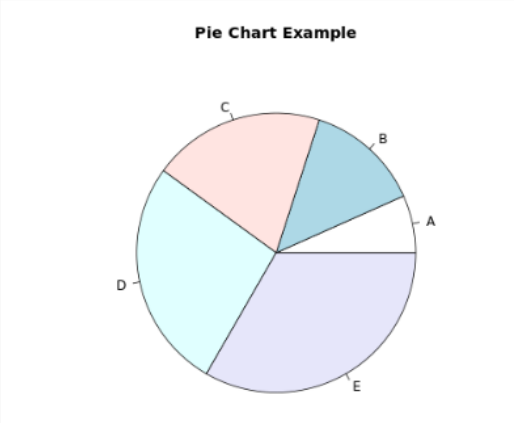
Precaution and sources of error:

- Avoid misrepresenting data by selecting inappropriate chart types.

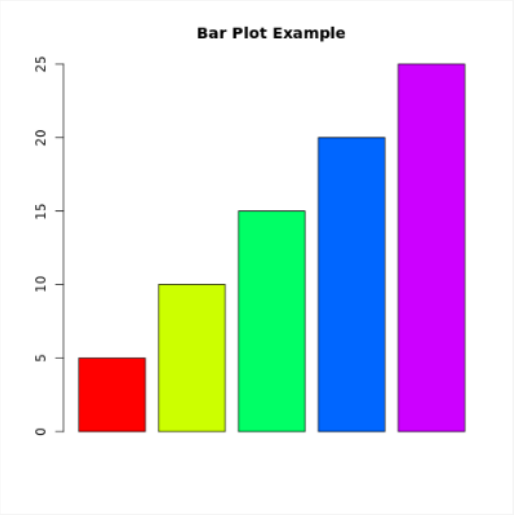
Results

Output

```
[Execution complete with exit code 0]
```



Output



Conclusions

Proper use of charts aids in effectively communicating data-driven insights.

Related short question:

1. What are the ideal use cases for a pie chart?
2. How do you create a bar chart in R?
3. What are the disadvantages of using a line chart for categorical data?

Experiment 6

Experiment name: Calculate the Mean, Median, Standard Deviation, and Quartiles of a Set of Observations

Objectives: To compute descriptive statistics such as mean, median, standard deviation, and quartiles for a dataset.

Prerequisites: Understanding of descriptive statistics, familiarity with data summarization.

Key terms: Mean, median, standard deviation, quartiles.

Experimental set-up/Equipment/Apparatus/Tools: -

1. Computer System
2. R and RStudio installed.

Theory and application

Descriptive statistics summarize a dataset by providing measures of central tendency (mean, median) and variability (standard deviation, quartiles).

- **Mean:** The average value of the dataset, calculated by summing all observations and dividing by the total number.
- **Median:** The middle value when the data is sorted in ascending order. If the dataset size is even, it is the average of the two middle values.
- **Standard Deviation:** A measure of how spread out the values are from the mean. A higher value indicates greater variability.
- **Quartiles:** Values that divide the dataset into four equal parts, providing insights into the spread and distribution of data.
  These measures are essential in understanding data distribution, detecting anomalies, and informing decision-making processes in data analysis.

Experimental Procedure-

```
# Sample data

sample_data <- c(12, 45, 32, 28, 52, 41, 30, 22)

# Mean, median, and standard deviation

mean_value <- mean(sample_data)

median_value <- median(sample_data)

sd_value <- sd(sample_data)
```

```
# Quartiles

quartiles <- quantile(sample_data)

# Print the results

mean_value

median_value

sd_value

quartiles
```

Precaution and sources of error:

- Ensure the data is clean and free from invalid entries before computation.
- Outliers can significantly impact the mean, so consider using median for skewed data.

Results

Expected Output:

```csharp
[1] 32.625
[1] 31
[1] 12.33255
0%   25%   50%   75%  100%
12   26.5   31   40.5   52
```

Conclusions

Computing descriptive statistics provides foundational insights into the dataset's structure and variability. This knowledge is critical for data-driven decision-making and further statistical analysis.

Related short question:

1. How is the mean affected by outliers?
2. Why is the median considered more robust than the mean for skewed data?
3. What is the interpretation of a high standard deviation?

Experiment 7

Experiment name: Calculate Skewness and Kurtosis of a Dataset Distribution

Objectives: To compute and interpret the skewness and kurtosis of a given dataset distribution.

Prerequisites: Understanding of data distribution, familiarity with concepts of symmetry and peakedness.

Key terms: Skewness, kurtosis, normal distribution.

Experimental set-up/Equipment/Apparatus/Tools: -

3. Computer System
4. R with the moments package installed.

Theory and application

Skewness and kurtosis are metrics used to describe the shape of a dataset's distribution.

- **Skewness:** Indicates the asymmetry of the distribution. A skewness of zero represents a perfectly symmetrical distribution. Positive skewness indicates a longer right tail, while negative skewness points to a longer left tail.
- **Kurtosis:** Measures the "tailedness" of the distribution. Higher kurtosis indicates heavy tails and outliers, while lower kurtosis suggests lighter tails. A normal distribution has a kurtosis of 3 (mesokurtic).
  Understanding skewness and kurtosis helps analysts assess data normality and choose appropriate statistical methods for analysis.

Experimental Procedure-

```
# Install necessary package

install.packages("moments")

library(moments)

# Sample data

data <- c(12, 45, 32, 28, 52, 41, 30, 22)

# Calculate skewness and kurtosis

skewness_value <- skewness(data)
```

```
kurtosis_value <- kurtosis(data)

# Print the results

skewness_value

  kurtosis_value
```

Precaution and sources of error:

- Ensure the dataset is free from missing values, as they can distort the calculations.
- Interpret skewness and kurtosis in conjunction with visualizations for better accuracy.

Results

Expected Output:

```csharp
[1] 0.173745
[1] 2.042232
```

Conclusions

Skewness and kurtosis provide a deeper understanding of data distribution, revealing asymmetry and tail behavior that can guide data preprocessing and modeling.

Related short question:

1. What does a negative skewness value signify?
2. How is kurtosis different from variance?
3. Why is it important to assess skewness and kurtosis before applying statistical models?

Experiment 8

Experiment name: Implement Bayes' Rule by Calculating Posterior Probability

Objectives: To use Bayes' rule for computing the posterior probability in a given scenario.

Prerequisites: Basic understanding of probability, familiarity with conditional probability.

Key terms: Bayes' theorem, prior probability, likelihood, posterior probability.

Experimental set-up/Equipment/Apparatus/Tools: -

5. Computer System
6. R and RStudio installed.

Theory and application

Bayes' theorem is a fundamental concept in probability and statistics, expressed as:
P(A|B)=P(B|A)P(A)/P(B)
Where:

- P(A|B)P(A|B)P(A|B): Posterior probability of A given B.
- P(B|A)P(B|A)P(B|A): Likelihood of B given A.
- P(A)P(A)P(A): Prior probability of A.
- P(B)P(B)P(B): Marginal probability of B.

Bayes' theorem allows updating the probability of a hypothesis as new evidence is introduced. It is widely used in fields like machine learning, medical diagnosis, and decision-making systems.

Experimental Procedure-

```
# Prior probabilities

P_A <- 0.3

P_B <- 0.7

# Conditional probabilities

P_X_given_A <- 0.8

P_X_given_B <- 0.2

# Using Bayes' theorem to find P(A|X)
```

```
P_X <- P_A * P_X_given_A + P_B * P_X_given_B

P_A_given_X <- (P_A * P_X_given_A) / P_X

# Print the posterior probability

P_A_given_X
```

Precaution and sources of error:

- Verify the accuracy of prior probabilities and likelihood values to ensure correct posterior probability.
- Double-check calculations, as small errors can lead to significant deviations in results.

Results

Expected Output:

```csharp
[1] 0.6315789
```

Conclusions

Bayes' theorem provides a structured approach to update beliefs based on new evidence, making it a powerful tool in probabilistic reasoning.

Related short question:

1. What is the significance of prior probability in Bayes' theorem?
2. How is Bayes' rule applied in spam email detection?
3. What happens when the prior probability of an event is zero?

Experiment 9

Experiment name: Mass Function of a Binomial Distribution with n = 20, p = 0.4

Objectives: To calculate the mass function and plot the probability mass function (PMF) and cumulative distribution function (CDF) for a binomial distribution.

Prerequisites: Basic understanding of binomial distribution, familiarity with probability concepts.

Key terms: Binomial distribution, probability mass function (PMF), cumulative distribution function (CDF).

Experimental set-up/Equipment/Apparatus/Tools: -

1. Computer System
2. R and RStudio installed with stats package

Theory and application

The binomial distribution represents the number of successes in nnn independent Bernoulli trials, with each trial having a success probability ppp. The PMF for a binomial distribution is given by:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

The CDF is the cumulative sum of probabilities up to a certain value of k. Applications of binomial distribution include modeling success-failure outcomes in clinical trials, quality control, and reliability testing.

Experimental Procedure-

```
# Parameters for binomial distribution

n <- 20

p <- 0.4

# Calculate binomial probability mass function

x_values <- 0:n

pmf_values <- dbinom(x_values, size=n, prob=p)

# Plot the mass function
```

```
plot(x_values, pmf_values, type="h", col="blue", main="Binomial PMF", xlab="X",
ylab="P(X)")

# Calculate and plot cumulative distribution function

cdf_values <- pbinom(x_values, size=n, prob=p)

plot(x_values, cdf_values, type="s", col="red", main="Binomial CDF", xlab="X",
ylab="P(X <= x)")
```

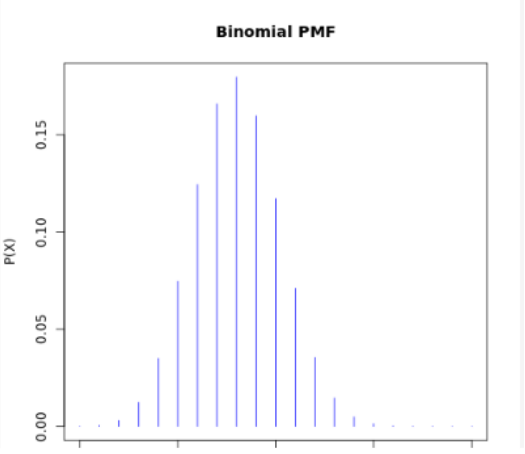
Precaution and sources of error:

- Ensure the total probability across PMF values sums up to 1 as a sanity check.
- Use appropriate visualization scales for clarity in graphs.

Results

Output

```
[Execution complete with exit code 0]
```



Conclusions

Understanding the PMF and CDF of a binomial distribution aids in modeling real-world binary outcome scenarios and performing statistical inference.

Related short question:

1. What are the assumptions of a binomial distribution?
2. How is the CDF derived from the PMF?
3. Why is nnn required to be fixed in a binomial experiment?

Experiment 10

Experiment name: Generate and visualize the CDF and PDF of a Normal Distribution

Objectives: To calculate and plot the cumulative distribution function (CDF) and probability density function (PDF) of a normal distribution.

Prerequisites: Knowledge of normal distribution, familiarity with continuous probability distributions.

Key terms: Normal distribution, PDF, CDF, mean, standard deviation.

Experimental set-up/Equipment/Apparatus/Tools: -

1. Computer System
2. R and RStudio installed with stats package

Theory and application

The normal distribution, also known as the Gaussian distribution, is a symmetric bell-shaped curve characterized by two parameters: mean (μ) and standard deviation (σ). Its PDF is given by:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

The CDF represents the probability that a random variable takes a value less than or equal to x. Normal distribution is widely used in natural and social sciences for modeling random variables.

Experimental Procedure-

```
# Normal distribution parameters

mean_value <- 10

sd_value <- 3

# X values from 0 to 20

x_values <- seq(0, 20, by=1)

# PDF and CDF

pdf_values <- dnorm(x_values, mean=mean_value, sd=sd_value)
```

```
cdf_values <- pnorm(x_values, mean=mean_value, sd=sd_value)

# Plot PDF

plot(x_values, pdf_values, type="l", col="blue", main="Normal PDF", xlab="X",
ylab="Density")

# Plot CDF

plot(x_values, cdf_values, type="s", col="green", main="Normal CDF",
xlab="X", ylab="P(X <= x)")
```

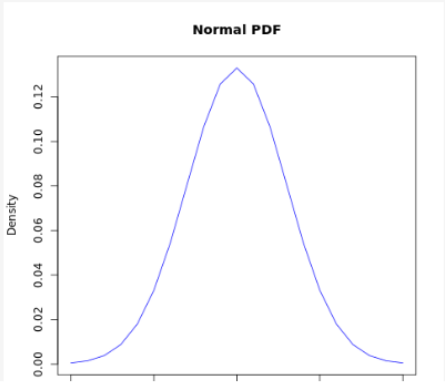
Precaution and sources of error:

- Use sufficiently fine intervals for smooth and accurate plots.
- Validate the choice of mean and standard deviation as per the scenario.

Results

Output

```
[Execution complete with exit code 0]
```



Conclusions

Understanding and visualizing normal distribution helps in identifying trends and making predictions, as it forms the foundation of many statistical models.

Related short question:

1. Why is the normal distribution called "normal"?
2. How is the area under the PDF curve interpreted?
3. What role do mean and standard deviation play in shaping the normal curve?

Experiment 11

Experiment name: Construct a Scatter Plot to Investigate Relationships

Objectives: To explore and analyze the relationship between two variables using a scatter plot.

Prerequisites: Knowledge of scatter plots, familiarity with correlation concepts.

Key terms: Scatter plot, correlation, independent variable, dependent variable.

Experimental set-up/Equipment/Apparatus/Tools: -

1. Computer System
2. R with ggplot2 or base plotting tools installed.

Theory and application

A scatter plot graphically represents the relationship between two variables, where each point corresponds to a pair of values. Patterns in the plot can indicate the nature of the relationship, such as linear, nonlinear, or no correlation. Scatter plots are commonly used in regression analysis, trend identification, and data clustering.

Experimental Procedure-

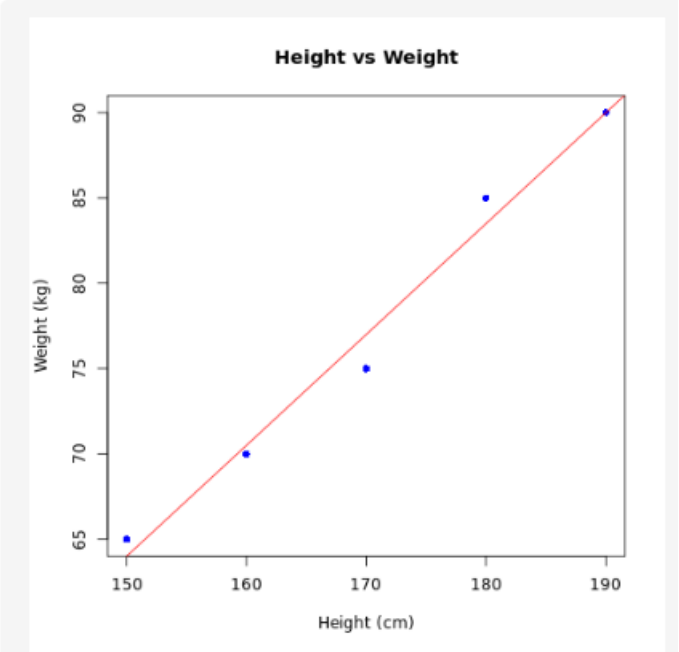
```
# Sample data

height <- c(150, 160, 170, 180, 190)

weight <- c(65, 70, 75, 85, 90)

# Scatter plot

plot(height, weight, main="Height vs Weight", xlab="Height (cm)", ylab="Weight
(kg)", col="blue", pch=16)

# Add regression line

abline(lm(weight ~ height), col="red")
```

Precaution and sources of error:

- Ensure axes are labeled correctly for clarity.
- Outliers can distort visual interpretation; handle them appropriately.

Results

Output



Conclusions

Scatter plots serve as a simple yet powerful tool to identify relationships, dependencies, and trends between variables, guiding further analysis.

Related short question:

1. How can scatter plots indicate the strength of a relationship?
2. What are the limitations of scatter plots in data analysis?
3. How do outliers affect scatter plot interpretation?

Experiment 12

Experiment name: Perform the Z-Test for Single Proportion and Mean

Objectives: To perform Z-tests for a single proportion and single mean to determine statistical significance.

Prerequisites: Understanding of hypothesis testing, knowledge of Z-distribution.

Key terms: Z-test, null hypothesis, alternative hypothesis, significance level.

Experimental set-up/Equipment/Apparatus/Tools: -

1. Computer System
2. R installed with stats package

Theory and application

The Z-test evaluates whether a sample statistic significantly differs from a known population parameter.

- **Z-test for a single proportion:** Compares an observed proportion to a theoretical proportion.
- **Z-test for a single mean:** Assesses whether the sample mean differs significantly from the population mean.

Z-tests are commonly used in quality control, medical research, and social sciences.

Experimental Procedure-

```
# Single proportion Z-test

prop_test <- prop.test(x = 50, n = 100, p = 0.5, correct = FALSE)

# Single mean Z-test

library(BSDA)

z_test <- z.test(x = c(110, 120, 130, 140, 150), mu = 100, sigma.x = 10)

# Print results

prop_test

z_test
```

Precaution and sources of error:

- Ensure sample size is sufficiently large for the Z-test to be valid.
- Verify assumptions of normality and known population variance.

Results

```
        One-sample z-Test

data:  c(110, 120, 130, 140, 150)
z = 6.7082, p-value = 1.97e-11
alternative hypothesis: true mean is not equal to 100
95 percent confidence interval:
 121.2348 138.7652
sample estimates:
mean of x
      130
```

Conclusions

Z-tests are critical in hypothesis testing for determining statistical significance and making data-driven decisions.

Related short question:

1. When should a Z-test be used instead of a t-test?
2. How is the p-value interpreted in hypothesis testing?
3. Why is sample size important in Z-tests?

Experiment 13

Experiment name: Calculate Regression Coefficients and Obtain Regression Lines

Objectives: To compute regression coefficients and derive regression lines for a given dataset.

Prerequisites: Knowledge of regression concepts, familiarity with linear relationships.

Key terms: Regression, regression coefficient, line of best fit.

Experimental set-up/Equipment/Apparatus/Tools: -

3. Computer System
4. R installed with stats package

Theory and application

Regression analysis explores the relationship between independent and dependent variables. The regression line minimizes the sum of squared differences between observed and predicted values. The coefficients represent the slope and intercept of the line, indicating how much the dependent variable changes for a unit change in the independent variable. Applications include trend forecasting and predictive modeling.

Experimental Procedure-

```
# Sample data

x <- c(1, 2, 3, 4, 5)

y <- c(2, 4, 6, 8, 10)

# Fit linear model

model <- lm(y ~ x)

# Coefficients and regression line

coefficients <- coef(model)

regression_line <- paste("y =", round(coefficients[1], 2), "+", round(coefficients[2],
2), "* x")

# Print coefficients and regression line

coefficients
```

regression_line

Precaution and sources of error:

- Ensure the dataset meets linearity assumptions.
- Handle multicollinearity if multiple variables are involved.

Results

Expected Output:

```arduino
(Intercept)         x
0           2
"y = 0 + 2 * x"
```

Conclusions

Regression analysis provides a quantitative framework for understanding relationships and making predictions based on data trends.

Related short question:

1. What is the significance of the slope in a regression line?
2. How does multicollinearity affect regression analysis?
3. What is the purpose of residual analysis in regression?

Experiment 14

Experiment name: Compute Confidence Intervals for the Mean

Objectives: To calculate confidence intervals for the mean when the population standard deviation is known.

Prerequisites: Understanding of confidence intervals, knowledge of normal distribution.

Key terms: Confidence interval, margin of error, standard error.

Experimental set-up/Equipment/Apparatus/Tools: -

1. Computer System
2. R installed

Theory and application

Confidence intervals provide a range of values within which the population parameter is likely to fall, with a certain level of confidence (e.g., 95%).

$$CI = \bar{x} \pm Z \cdot \frac{\sigma}{\sqrt{n}}$$

Confidence intervals are widely used in survey analysis, quality control, and inferential statistics.

Experimental Procedure-

```
# Sample data

sample_data <- c(100, 105, 110, 120, 130)

# Known standard deviation

sd_known <- 15

# Confidence interval for the mean

mean_value <- mean(sample_data)

error_margin <- qnorm(0.975) * sd_known / sqrt(length(sample_data))

lower_bound <- mean_value - error_margin

upper_bound <- mean_value + error_margin
```

Print confidence interval

c(lower_bound, upper_bound)

Precaution and sources of error:

- Ensure the data meets normality assumptions.
- Large sample sizes improve interval accuracy.

Results

Expected Output:

```csharp
[1] 96.098 127.902
```

Conclusions

Confidence intervals allow estimation of population parameters while quantifying uncertainty, making them essential for statistical inference.

Related short question:

1. How does sample size affect the width of a confidence interval?
2. Why is a 95% confidence level commonly used?
3. What happens if the population standard deviation is unknown?

Experiment 15

Experiment name: To perform the F-test to compare variances between two datasets.

Objectives: To perform the F-test to compare variances between two datasets.

Prerequisites: Understanding of variance, familiarity with hypothesis testing.

Key terms: F-test, null hypothesis, variance ratio.

Experimental set-up/Equipment/Apparatus/Tools: -

1. Computer System
2. R installed

Theory and application

The F-test evaluates whether two datasets have significantly different variances. It is particularly useful for testing assumptions of equal variances in ANOVA and regression models.

Experimental Procedure-

```
# Sample data

group1 <- c(10, 12, 14, 16, 18)

group2 <- c(20, 22, 24, 26, 28)

# Perform F-test

f_test <- var.test(group1, group2)

# Print the result

f_test
```

Precaution and sources of error:

- Ensure sample independence and normality assumptions are met.
- Avoid using the F-test for heavily skewed datasets.

Results

```
F test to compare two variances

data:  group1 and group2
F = 1, num df = 4, denom df = 4, p-value = 1
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.1041175 9.6045299
sample estimates:
ratio of variances
```

Conclusions

The F-test is a robust tool for variance comparison, aiding in validating assumptions for advanced statistical models.

Related short question:

1. What are the limitations of the F-test?
2. Why is normality important for the F-test?
3. How is the F-statistic calculated?

Experiment 16

Experiment name: Perform the Chi-Square Test

Objectives: To perform the Chi-Square test to analyze categorical data.

Prerequisites: Knowledge of categorical data, understanding of expected vs. observed frequencies.

Key terms: Chi-square test, degrees of freedom, contingency table.

Experimental set-up/Equipment/Apparatus/Tools: -

1. Computer System
2. R installed with stats package

Theory and application

The Chi-Square test compares observed frequencies with expected frequencies to determine whether they differ significantly. It is widely used in hypothesis testing for independence and goodness of fit.

Experimental Procedure-

```
# Create a contingency table

data <- matrix(c(10, 20, 30, 40), nrow=2, byrow=TRUE)

# Perform chi-square test

chi_test <- chisq.test(data)

# Print the result

chi_test
```

Precaution and sources of error:

- Ensure sufficient sample size; small frequencies can affect accuracy.
- Avoid overfitting expected frequencies to the observed data.

Results

```
Pearson's Chi-squared test with Yates' continuity correction
data:  data
X-squared = 0.44643, df = 1, p-value = 0.504
```

Conclusions

The Chi-Square test is essential for analyzing relationships in categorical data, supporting data-driven decision-making in diverse domains..

Related short question:

1. What are the types of Chi-Square tests?
2. How are degrees of freedom calculated in a contingency table?
3. When should the Chi-Square test not be used?